



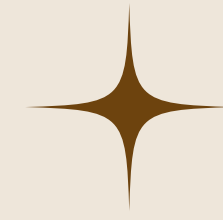
PETRA

Cloud-Based Petition Management
and Monitoring System

TEAM

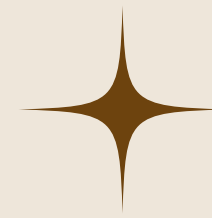
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PROBLEM STATEMENT



Existing petition and complaint management systems are manual, inefficient, and lack a centralized platform for submitting requests with documents like PDFs or audio. Users face difficulties in tracking status, while administrators struggle to handle large volumes due to lack of automation, intelligent classification, and priority-based handling, where critical issues (e.g., healthcare) should be addressed before less urgent ones (e.g., street lighting). These systems also lack secure authentication, real-time notifications, data-driven insights, and efficient background processing, leading to delays, mismanagement, reduced transparency, poor decision-making, and ultimately low user satisfaction and operational inefficiencies.

PROPOSED SOLUTION



Cloud-based AI-powered system providing a centralized platform for submitting and managing petitions with PDF/audio support.



NLP-based classification and rule-based priority scoring to categorize and prioritize critical issues effectively.



Secure authentication, real-time tracking, notifications, and scalable Azure cloud infrastructure for efficient processing.

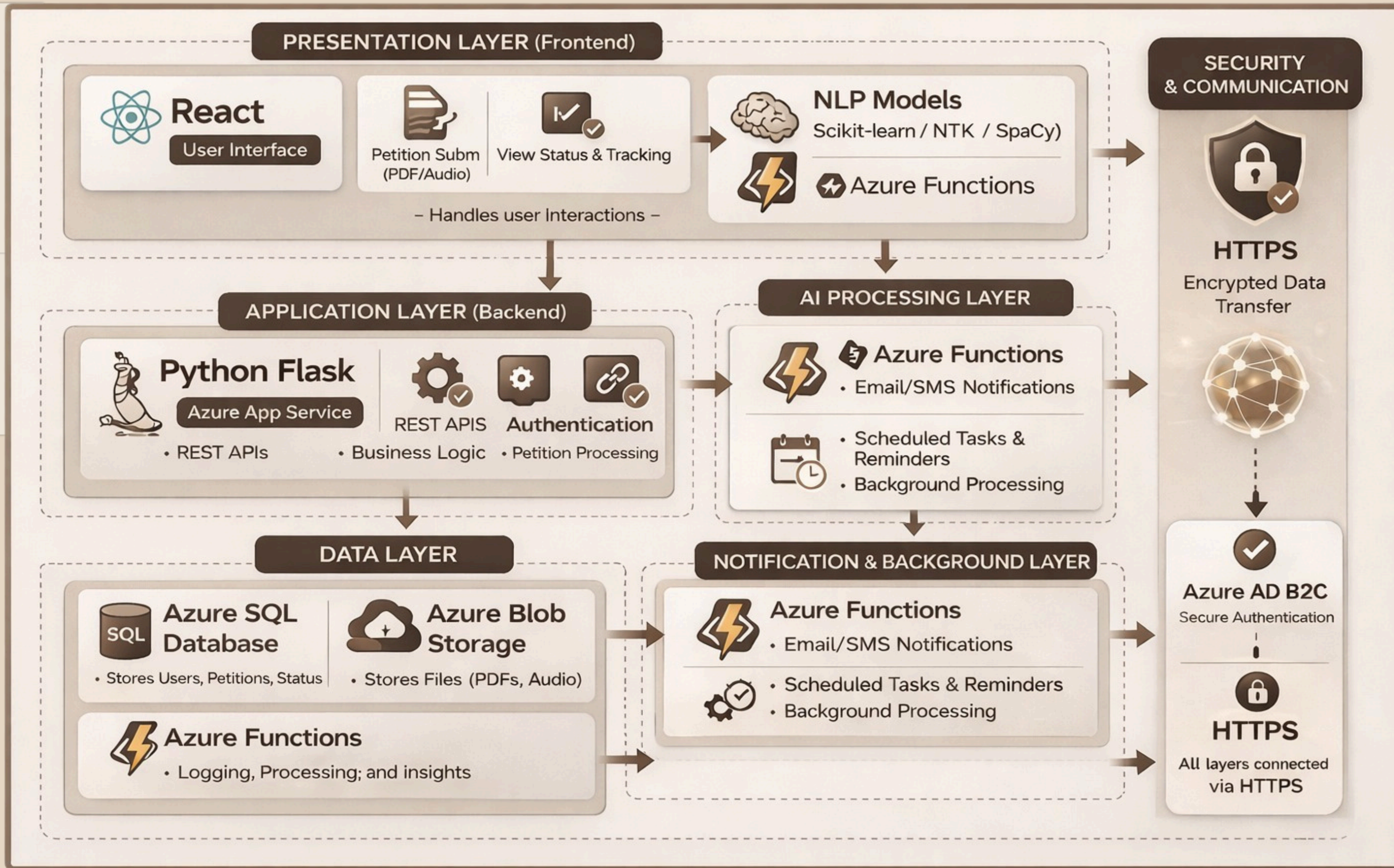
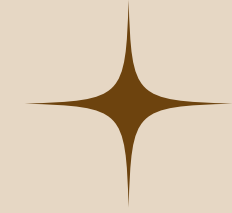


Analytics and monitoring tools to enable data-driven decisions, improving transparency, accountability, and user satisfaction

ARCHITECTURE MODEL

- 1 **Public cloud deployment using Microsoft Azure, built on Platform as a Service (PaaS) for scalability and flexibility.**
- 2 **Azure App Service hosts the Flask-based backend and APIs, while Flutter/React frontend enables user interaction and petition management.**
- 3 **Azure Functions handle serverless processing, including AI-based petition classification, background jobs, and automated notifications (Email/SMS).**
- 4 **Azure SQL Database manages structured data (users, petitions, status), while Azure Blob Storage securely stores files like PDFs and audio.**
- 5 **Azure AD B2C ensures secure authentication and user identity management across the platform.**
- 6 **Power BI Embedded provides real-time analytics and insights, supported by Application Insights for monitoring and performance tracking.**
- 7 **Deployed in Central India region ensuring low latency, high availability, and a scalable, cost-efficient system.**

SYSTEM DESIGN





RELIABILITY & OPERATIONAL EXCELLENCE

Reliability

- **High availability using Azure cloud infrastructure**
- **Scalable system to handle large workloads**
- **Data backup and recovery with Azure SQL & Blob Storage**
- **Fault-tolerant design using Azure Functions**
- **Real-time monitoring with Application Insights**

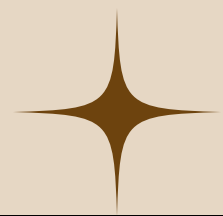
Operational Excellence

- **Automated workflows for petition processing and prioritization**
- **Efficient background processing using Azure Functions**
- **Continuous monitoring and logging for performance optimization**
- **Real-time notifications for users and admins**
- **Data-driven decision-making using Power BI**

FUNCTIONAL REQUIREMENTS



Feature	Expected Usage	Azure Spec	Scaling Method
User Interface (React)	Concurrent users submitting & tracking petitions	Azure App Service (Basic/Standard Plan)	Auto-scale based on CPU & requests
API & Backend (Flask)	Handle API calls, business logic, authentication	Azure App Service (Standard S1/S2)	Horizontal scaling (increase instances)
AI Classification & Priority	Process petition text for classification & priority	Azure Functions (Consumption Plan)	Event-driven auto-scaling
Database (Azure SQL)	Store users, petitions, status, logs	Azure SQL Database (DTU-based / vCore)	Vertical scaling (increase DTU/vCore)
File Storage (PDF/Audio)	Store uploaded documents & media files	Azure Blob Storage (Hot Tier)	Auto-scale storage capacity
Authentication	Secure login and user management	Azure AD B2C	Managed scaling (Azure handles automatically)
Notifications (Email/SMS)	Send alerts and updates to users/admins	Azure Functions + (Optional Twilio)	Event-based scaling
Background Processing	Scheduled tasks, reminders, queue handling	Azure Functions / Queue Storage	Auto-scale based on triggers
Analytics	Dashboard and reporting insights	Power BI Embedded	Scale via capacity tiers
Monitoring & Logging	Track performance and errors	Azure Application Insights	Auto-scale with app services



NON-FUNCTIONAL REQUIREMENTS

NFR Metric	Target	Azure Setup	Reason
Availability	≥ 99.9% uptime	Azure App Service SLA	Ensures system is always accessible
Performance	< 2 sec response time	App Service + (Optional CDN)	Fast and smooth user experience
Scalability	1000+ concurrent users	Auto-scale (App Service & Functions)	Handles increasing workload efficiently
Security	High data protection	Azure AD B2C + HTTPS + Encryption	Protects user data and privacy
Data Durability	99.999% reliability	Azure SQL + Blob Storage backups	Prevents data loss
Reliability	Fault-tolerant system	Azure Functions + Application Insights	Ensures continuous operation
Maintainability	Easy updates	Azure DevOps + Monitoring tools	Simplifies maintenance and debugging
Usability	User-friendly UI	React Frontend	Improves user satisfaction
Notification Latency	Near real-time	Azure Functions (event-triggered)	Ensures timely alerts
Analytics Accuracy	High accuracy	Power BI Embedded	Supports better decision-making

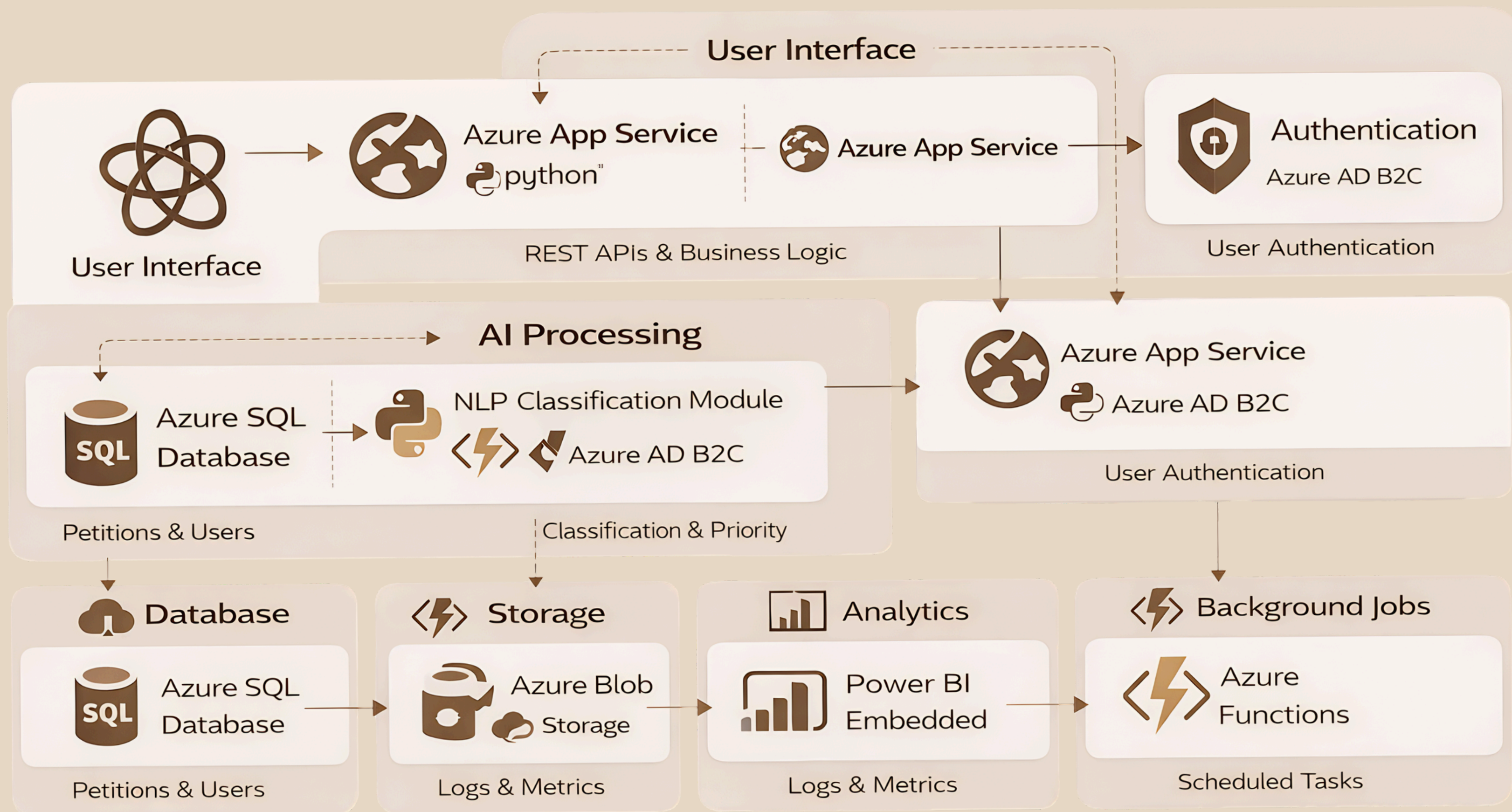
AZURE SERVICES MAPPING



Requirement	Azure Service	Tier / Plan	Purpose	Cost (₹/month approx.)
Frontend & Backend	Azure App Service	Runs frontend & backend apps	Hosts React UI and Flask APIs	₹1,500 – ₹4,000
Database	Azure SQL Database	Stores user & petition data	Maintains structured data	₹1,000 – ₹3,000
File Storage	Azure Blob Storage	Stores files (PDFs, audio)	Handles document uploads	₹200 – ₹800
Authentication	Azure AD B2C	User authentication system	Manages login and user identity	Free – ₹500
AI Processing	Azure Functions	Processes AI tasks	Handles classification & priority scoring	₹200 – ₹1,000
Notifications	Azure Functions + Twilio	Sends alerts & updates	Email/SMS notifications	₹300 – ₹1,500
Background Tasks	Azure Functions	Handles scheduled tasks	Reminders and background jobs	₹200 – ₹800
Analytics	Power BI Embedded	Shows reports & dashboards	Provides insights and analytics	₹5,000 – ₹15,000
Monitoring	Application Insights	Checks errors & performance	Tracks logs and system health	₹200 – ₹1,000
Backup	Azure Backup / Storage	Saves copy of data	Data recovery and backup	₹300 – ₹1,000



ARCHITECTURE MODEL





THANK YOU